

A simple 3.4GHz transverter



A Geelong Amateur Radio Club Project

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Lou Blasco VK3ALB – Documentation

David VK3HZ, Lee VK3PK, Tim VK5ZT – Software & coding

Introduction

The 3.4GHz band in Australia is a bit of a Cinderella band. Surplus equipment for this band has not been easy to find. Couple this with rumored closure of this band due to the roll out of the National Broadband network (NBN) and it's easy to understand the reluctance of amateurs to spend time, effort and money to get on the 3.4GHz band. Notwithstanding the above, the 3.4GHz band is still available to Australian amateurs and is an excellent band on which to experiment with microwaves.

The criteria for successful modification and take-up of surplus equipment for use in the 3.4GHz band are based on a number of things;

- Ease of access to suitable surplus equipment
- Ease of conversion without specialized test equipment
- Low entry cost – under \$100

This article deals with the conversion of surplus 3.5 GHz subscriber transceivers to 3.4GHz transverters. This transceiver meets the above criteria.

Amateurs wanting to experiment with these transceivers will need to undertake the following tasks.

- Understand transverter block diagrams
- Understand basic alignment techniques
- Read and understand simple circuits
- Remove two metal shields from the PCB.
- Remove/replace some surface mount and thru hole components
- Identify key locations on the PCB to connect wires and coax.
- Make some small PCB's using paddy board style construction
- Source simple electronic components to make the boards

The small boards are soldered to the main PCB to conserve space. Use a business card to space the boards clear of the base plate and make sure they are no wider than 32mm or they will not fit under the cover.

Don't forget to check component and board clearances as you assemble the various sections. It is very easy to foul the small boards or the large filter capacitor on the main PCB if you do not take care when positioning connectors on the back panel.

You will need a low power 70cm transceiver or hand held capable of transmitting on 444MHz to complete the installation. Whilst SSB transmissions certainly mean longer distance contacts there is no reason that a FM transceiver could not be used.

We recommend against using a high power radio turned down to the lowest output power. Whilst many people do this with success there is a lot of documentary evidence that shows most modern radios transmit a spike of RF power in some cases close to maximum output when first keyed. This happens even if the radio is adjusted for low power.

You will have to make some keying arrangement such as a bias tee, RF sense or external PTT line from your radio.

The conversion of the panel should take around 8 hours to complete and cost significantly less than \$50 in parts not including the panels.

These panels have been tested at 3398MHz and 3395MHz by simply changing the frequency on the IF radio. No significant degradation in performance was found at 3398 but at 3395 the TX output power had reduced by around 2.5dB.

For those wanting to try the microwave bands without breaking the bank this project certainly fills the bill.

The reader is encouraged to carefully study this document and understand the conversion process before commencing the work.

While it may seem there is repetition in some areas the idea is to reinforce the process by going into greater detail in each section.

Some assumptions have been made regarding technical and mechanical skills of the constructor but we acknowledge that we are not all the same.

The most difficult part for some will be removing the shields, for others it will be the external boards and for others perhaps the alignment process.

We are always happy to answer any questions you may have regarding this project.

Parts List

The parts required to complete the conversion is quite low and could feasibly be found in a well stocked junk box. All the parts can be purchased locally.

6 x 0.001uf SMD	3 x 6k8 resistors
1 x 2k2 0805 SMD	1 x 18R 1W resistor
2 x 56R 0805 SMD	2 x 100R 1W resistor
1 x 330R 0805 SMD	1 x 330R 1W resistors
2 x 200 ohm trimpots	2 x 560R 1W resistors
1 x 68R resistor	5 x 1N4148 switching diodes
1 x 680R resistor	1 x IN5401 power diode (optional)
1 x 820R resistor	2 x BC327 (or similar) transistors
1 x 1k0 resistor	1 x BC547 (or similar) transistor
1 x 4k7 resistor	

At least one TRI3500 panel transceiver (see conversion summary)
 1 x blank double sided PCB for IF board (30mm x 85mm)
 1 x blank double sided PCB for DC decode board (30mm x 40mm)

Length of RG316 or similar thin coax
 Several different colors of hookup wire
 RF changeover relay *
 IF changeover relay (RL1) Mini 12V DPDT Omron G2VN234P or similar **
 1 x BNC socket
 1 x Power socket
 RCA socket (for external PTT input – if required)

* Mini Kits stock a RF relay kit that is suitable for use in this project. However this kit also includes three SMA sockets which might be difficult to fit in the available space. The better and cheaper option might be to order the relay and PCB separately and not use the SMA connectors. This will be the single most expensive part of the conversion project so good scrounging here could lead to significant savings.

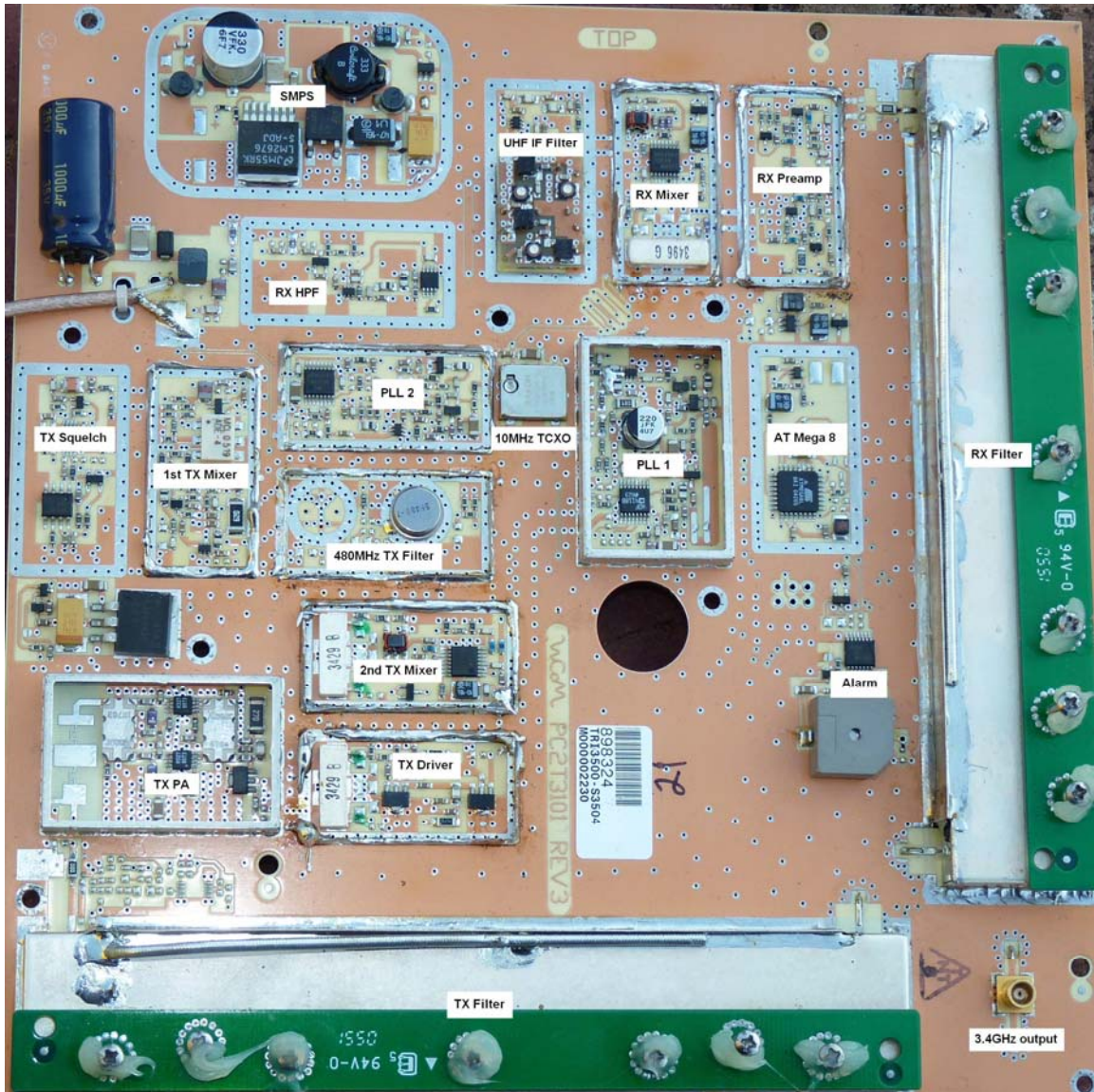
1 x RF Relay Kit - EME144-KIT or
 1 x Relay - G6Z-1F-A-DC5
 1 x PCB - EME144-PCB

Other suitable microwave relays can be found online.

** Relays from FM828s or old commercial sets can also be used for the IF change-over relay.

Main board layout

INFORMATION ONLY - DO NOT REMOVE ALL THE SHIELDS DURING CONVERSION WORK.



Some boards may vary slightly in appearance but all follow this general layout.

Conversion summary

The TRI3500 is a self contained full duplex transceiver operating on 3434MHz to 3442MHz for TX and 3484MHz to 3492MHz for RX. Included in the panel is a 3.4GHz patch antenna with a nominal gain of 18dBi. The transmitter is rated at +22dBm and the RX noise figure < 7dB. Whilst the label on the back of the panel specifies the supply voltage as +21V to +28V the converted panel works quite well on +12V DC.

The modification involves removing and/or exchanging some parts on the main PCB, adding wires and coax connections as well as making a couple of small boards to manage TX/RX switching as well as TX/RX enable.

The shields are removed and discarded from the 1st RX mixer and 480MHz filter. There is no need to remove any other shields.

The ceramic filter in the 1st RX mixer is replaced by the large TX filter.

Two coils in the UHF filter block are replaced with some home brew types.

The 1st TX mixer and 480MHz filter are bypassed and the 2nd TX mixer input attenuator is increased from 2dB to approximately 24dB.

A modification is made to the transverter TX Squelch line to permit external control of the PA.

The input diplexer is bypassed and the signal from the IF radio is switched to the TX or RX chain via an external IF Board.

An external DC enable board controls the IF and RF switching as well as switching the transverter from RX to TX and vica verca.

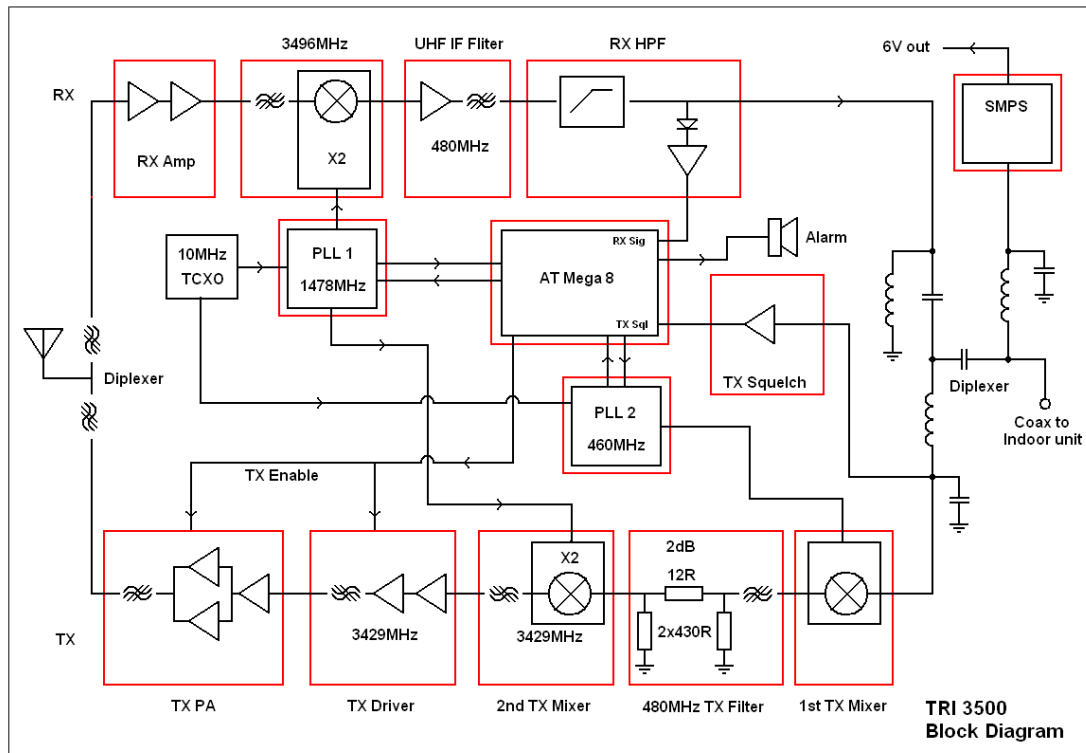
An external RF relay board switches the 3.4GHz RX/TX signals to the antenna.

After the main board has been modified a preliminary check of the board operation is undertaken.

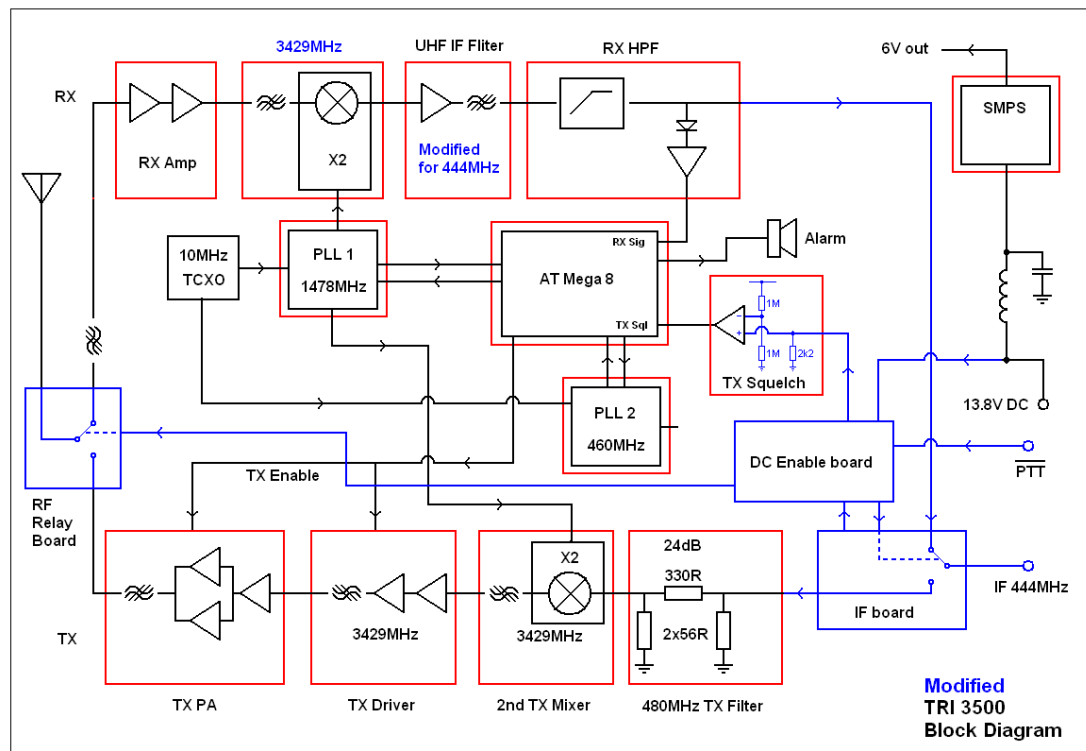
If testing is successful then the extra boards are installed and the remaining wiring is completed.

One final check is made to confirm everything is ok and the panel is ready for operation.

Block Diagrams



Unmodified unit



Modified unit

Conversion steps in detail

Before starting modifications give the board a visual inspection to ensure there is no obvious damage and check it is functional by applying 12VDC across the filter capacitor on the main PCB. You should hear beeping. A continuous series of dits is a good sign.

Start with all the steps that take a lot of heat. Don't try and force things rather use enough heat to make the solder flow freely. Make a jig to hold the board so you don't burn yourself or drop the board at the wrong time. Take care not to bump the board when the solder is flowing as you could move components or create shorts.

Remove the shields of 1st RX mixer and 480MHz filter blocks. They do not need to be replaced afterwards.

Be careful that the solder is completely molten before you attempt to remove a shield. It is very easy to damage the PCB coating or tracks and make sure you don't dislodge any components during this phase.

A technique that worked for the author was to heat the underside of the board beneath the shields with a heat gun. The heat transfers up through the PCB vias and softens the solder holding the shields. Then with another heat gun or heavy soldering iron heat the shields from the top until the solder is fully melted. Lift the shields straight up.

In the 1st RX mixer block either remove the 3496 filter or cut the leads off close to the body. You will be connecting thin coax lines from the pads that the filter was connected to the TX filter. See detailed image of the TX filter installation

In the 480MHz filter block, remove and discard the round filter and replace the attenuator with 2 x 56 ohm and 1 x 330 ohm resistors. (~24dB) See detailed image of the 480MHz filter block.

On the underside of the board completely remove the short TX cable that goes from the output connector to the TX filter.

On the underside of the board disconnect the short RX cable at the end closest to the MCX connector. Do not disconnect the other end.

Remove the MCX socket from the top of the board and keep it for use with the MiniKits relay board later on.

In the 1st RX mixer, using thin PTFE coax (RG316 or similar) connect the TX mixer in place if the 3496 filter. See detailed image of the TX filter installation.

Remove the zero ohm resistor from the output of the PA.

Modify the TX squelch block in preparation for the TX enable signal. See detailed images of the TX Squelch block.

Remove the two outer coils on the UHF IF filter board and replace them with two 3T coils wound on a 2mm former. The exact diameter of the wire is not critical but should be around 0.5mm diameter. Do not touch the centre coil. See detailed image of the RX filter modification.

Remove the two SMD inductors from the original IF input diplexer. Leave all the capacitors in place. See detailed image of the input section.

Build & test the IF, DC Enable and RF Relay boards but do not attach them to the main board right away.

Testing of the main board is done before the external boards and final wiring are installed. In this way if issues are found it is still quite early in the job and there are fewer possibilities for having introduced a problem.

Depending on the amount of test equipment you have the testing could take some time. If you follow the instructions in this manual you will need to attach coax leads with appropriate connectors to the 70cm RX and TX ports as well as the 3.4GHz RX & TX ports.

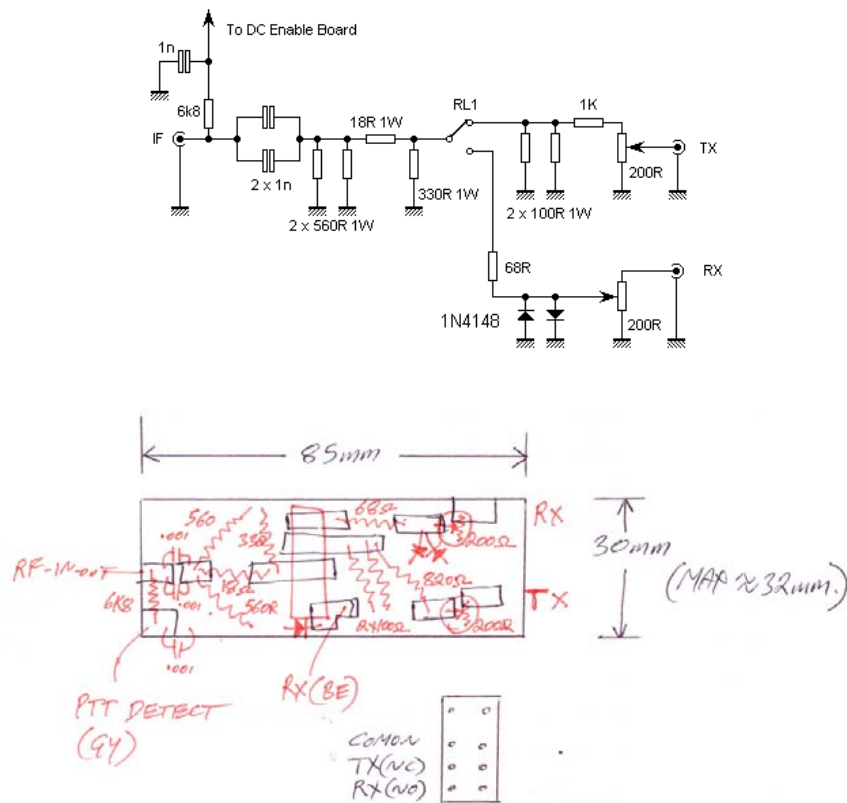
Once the initial testing is completed the external boards and wiring are added and a final check is made to ensure the transverter is ready for action.

Review the pictures of the completed transverter and mark out where you will mount the boards.

IF board details

The IF board detects PTT from the radio and passes it to the DC enable board. Your radio must put DC on the RF line or you have to build a bias tee or sense the presence of RF in some way. The RF to/from the radio is switched to the transverter via RL1. The TX and RX lines go to the transverter board and bypass the original input diplexer.

Expect an input VSWR of around 2:1 for this board.



The PCB is made from double sided blank PCB. It should be no wider than 32mm or it will not fit inside the back cover. The tracks are made by cutting copper away from the board using a sharp hobby knife. Use this diagram and the close-up pictures as well as the circuit to confirm component locations. It should be obvious that the backplane must be grounded using good RF techniques.

Once completed, test the board on the bench before soldering it onto the main PCB.

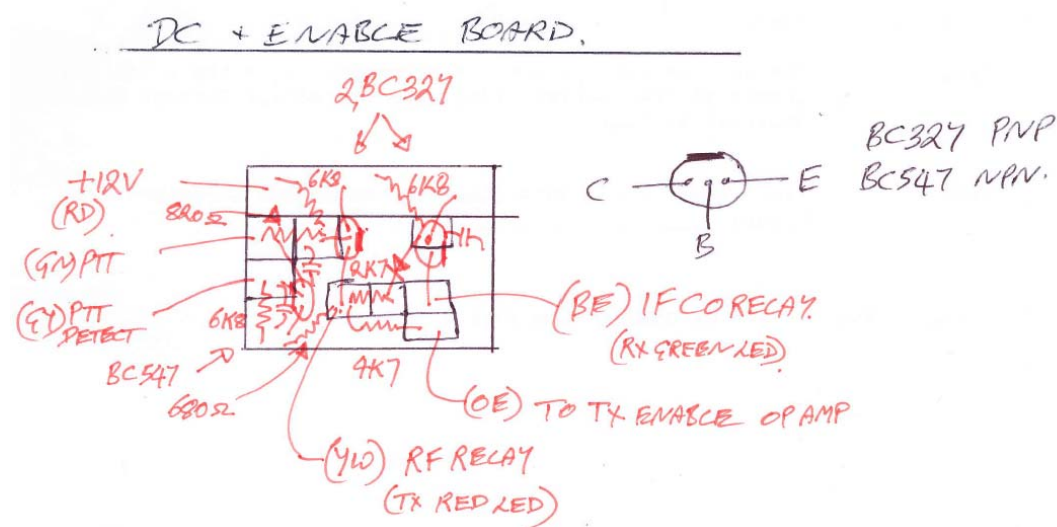
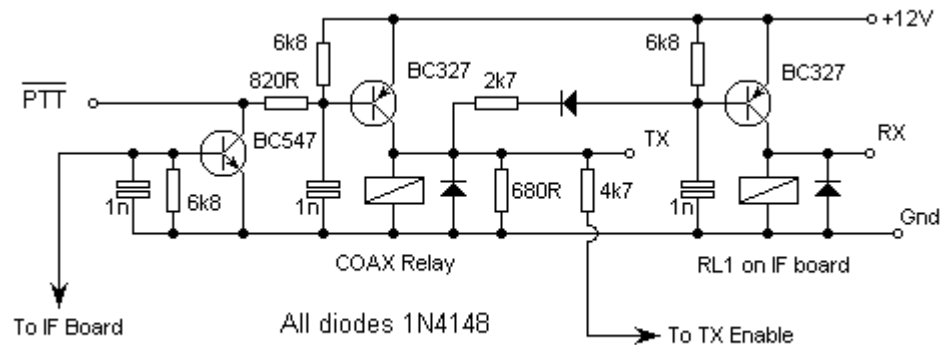
The board is designed for 2.5W but should comfortably handle 5W SSB. We recommend the attenuator resistors be increased to a higher power rating if FM or higher power operation is anticipated.

DC enable board details

The DC enable board takes the PTT detect line from the IF board and uses it to switch the transverter from TX to RX and back. It is also possible to use an external PTT by pulling the collector of the BC547 to ground.

The TX enable line goes to the transverter board

The TX & RX connections can driver LEDs through suitable dropping resistors.



The PCB is made from double sided blank PCB. It should be no wider than 32mm or it will not fit inside the back cover. The tracks are made by cutting copper away from the board using a sharp hobby knife. Use this diagram and the close-up pictures as well as the circuit to confirm component locations. It should be obvious that the backplane must be grounded using good RF techniques.

Once completed, test the board on the bench before soldering it onto the main PCB.

Transverter conversion checklist

Refer back to the previous section(s) and the pictures section if you are unclear about any of these steps.

- ☐ Build & test IF, DC and relay boards and put them to one side
- ☐ Initial testing of main board successful?
- ☐ Remove the shields for the 1st RX Mixer and 480MHz TX filter.
- ☐ Remove or disconnect the ceramic filter marked 3496 from the RX mixer.
- ☐ Remove the round filter can marked SF480 from the 480MHz TX filter.
- ☐ Replace attenuator in the 480MHz TX filter block
- ☐ On the underside of the PCB remove the TX cable completely.
- ☐ On the underside of the PCB unsolder the RX coax tail closest to the MCX connector.
- ☐ Remove the MCX output connector and solder to the common terminal (underside) of the MiniKits RF relay board.
- ☐ Using thin PTFE coax connect the TX filter into the 1st RX mixer.
- ☐ Remove the zero ohm resistor between the TX PA and large TX filter.
- ☐ Modify TX Squelch block
- ☐ Remove the two SMD RF chokes at the IF coax input.
- ☐ Replace input & output coils on UHF IF filter board
- ☐ Attach DC input leads directly across the 1000uf electro.
- ☐ Add reverse voltage protection diode at the power input. (Optional)

- ☐ Go to the testing page and check the board works after the modifications.

- ☐ Only continue with the next steps if testing was successful.

- ☐ Attach the IF, DC enable and relay boards to the main board.
- ☐ Connect a thin coax from the input of the 24dB attenuator in the 480MHz TX filter to the IF board.
- ☐ Connect a thin coax from output of the RX HPF block to the IF board.
- ☐ Connect a thin coax from the output of the TX PA to the RF relay board.
- ☐ Connect the previously disconnected coax tail from the RX filter to the RF relay board.
- ☐ Connect a hookup wire from the TX enable input to the DC enable board.
- ☐ Connect a hookup wire from the +12V input to the DC enable board.
- ☐ Connect a hookup wire from the DC enable board to the coax relay board.
- ☐ Link the DC enable board to the IF board (refer circuits).
- ☐ Check your work and make sure you have everything right.

- ☐ Go back to the testing page and complete the transverter adjustments.

OK, that's it. You should now have a working transverter.

Testing

Perform these tests before connecting the IF, DC & relay boards to the main PCB.

TX Testing

- Connect a power meter (1W @3.4GHz) directly to the output of the TX.
- Connect a 70cm signal generator to the TX input.
- Apply +12V to the panel, supply current should be ~ 400mA
- Connect TX enable to +12V via a 4k7 resistor, supply current should rise to ~ 800mA
- Apply 444MHz ~ -4dBm to TX input.
- TX output @ 3400MHz should be around +28 to +29dBm
- Vary the input level and check the output changes in a linear fashion.

RX Testing

- Screw in all screws of **both filters** except for those second from each end.
- Screw out the second screws from each end almost all the way out.
- Connect a 444MHz RX to the board.
- Apply -50dBm @ 3400MHz to the transverter RX input.**
- Adjust all screws that were initially screwed in for best RX.
- Remove 3400MHz signal and switch off RX AGC on radio if possible.
- Adjust the middle trimmer cap on UHF filter for max audio noise.
- Flex remaining two RX coils and re-peak middle trimmer cap.
- Now adjust the remaining two trimmer caps, output trimmer is quite broad.
- A reasonable increase in RX noise should occur when the main board is switched on, 15-20dB increase is good.

Now complete the installation of the IF, DC & relay boards and wiring of the panel then return here for final adjustments of the IF board.

After final assembly repeat the RX alignment with a lower input signal. The RX pot should be set at or near maximum with no preamp, but S meter should not move on RX when panel is switched on.

Set TX drive pot to mid travel or ground end, connect a suitable microwave power meter to the relay MCX output connector and apply carrier from your IF rig and wind up the drive pot until the output is a little below what you measured without the relay board. Halve the IF drive if convenient and observe around 1.5 to 2dB drop in output.

Netting (if required) can be done by carefully adjusting the 10MHz TCXO.

** For those without a signal source see the procedure on the following page.

Adjusting the receiver by beeper

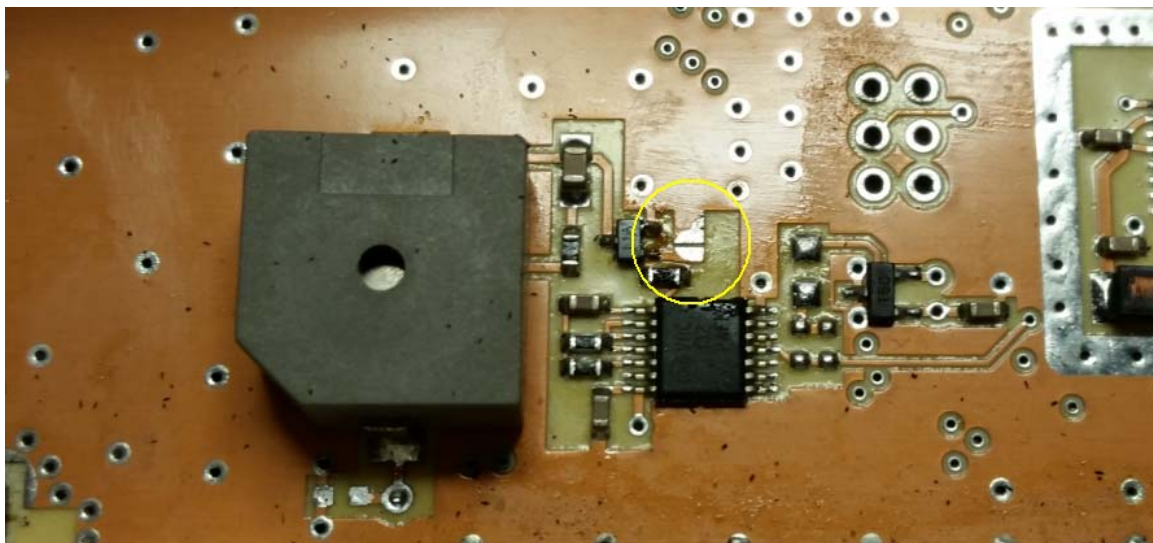
The home builder with little specialist test equipment can align these panels using the internal audible S meter to peak the 3.4GHz RX filter and the UHF IF filter coils & trimmers. This process presumes you have some kind of 3.4GHz signal source or an extra transmitting panel

The panel contains a beeper which is used as an audible S meter by the installation technicians to find the correct bearing to the base transmitter. The stronger the received signal, the faster the beeps. The audible S meter is quite responsive to changes in signal strength. In normal operation the beeper emits 2000 pulses then stops. The faster the beeps the shorter the time the beeper runs. Every time the power is cycled the beeper runs for another 2000 cycles.

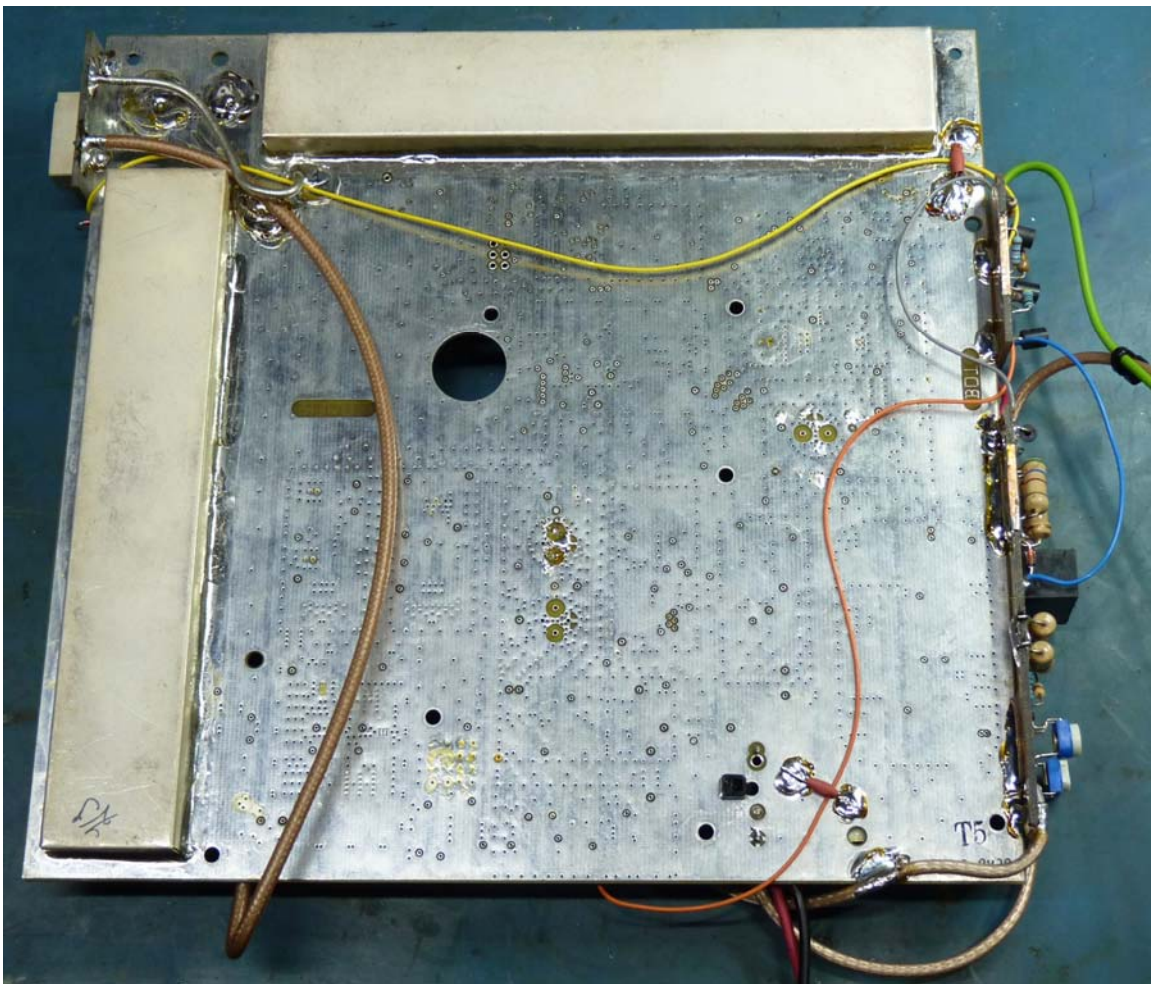
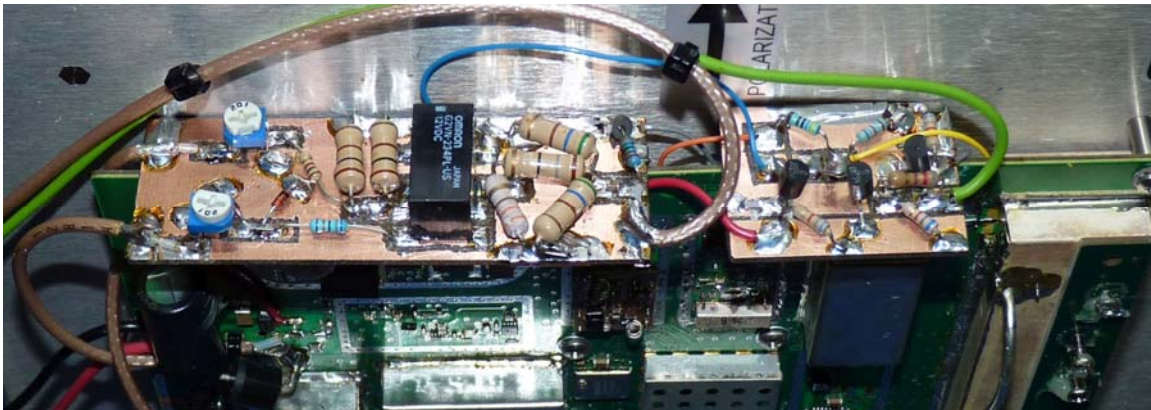
When the panel senses the RF it will start beeping at a faster rate. Set another panel up on transmit at a distance where the beeps have just started to speed up on the panel you are tuning. Continue with the alignment as outlined in the tune up notes adjusting the filter/coils to increase the repetition rate. Once the repetition rate plateaus increase the distance between the panels and continue the adjustments to see if you can get any more out of it.

Once you have finished, you can disable the beeper by putting a blob of solder across the pads indicated in the following image.

Go back and complete the adjustment of the UHF filter as detailed on the previous page.

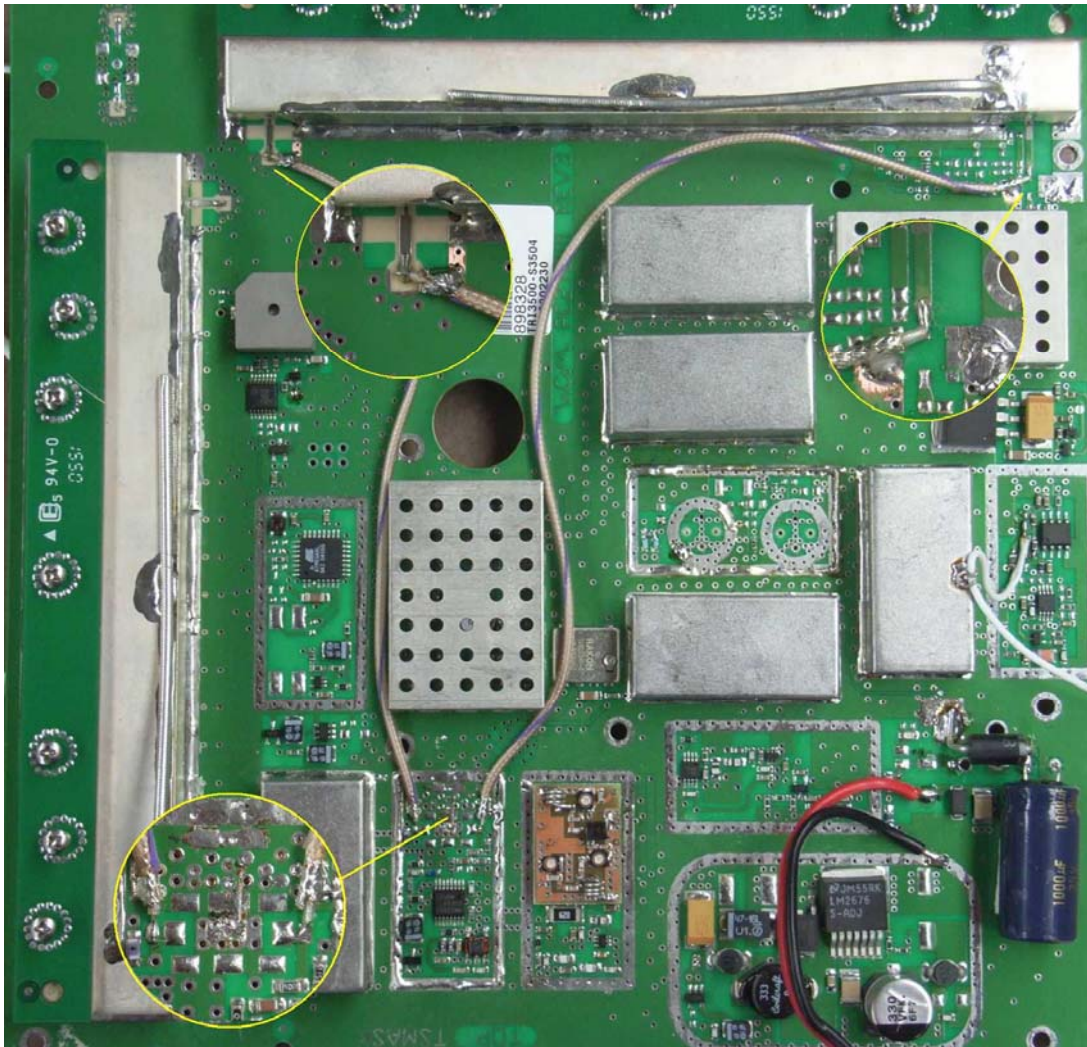


PCB pictures



Underside of board

Filter wiring



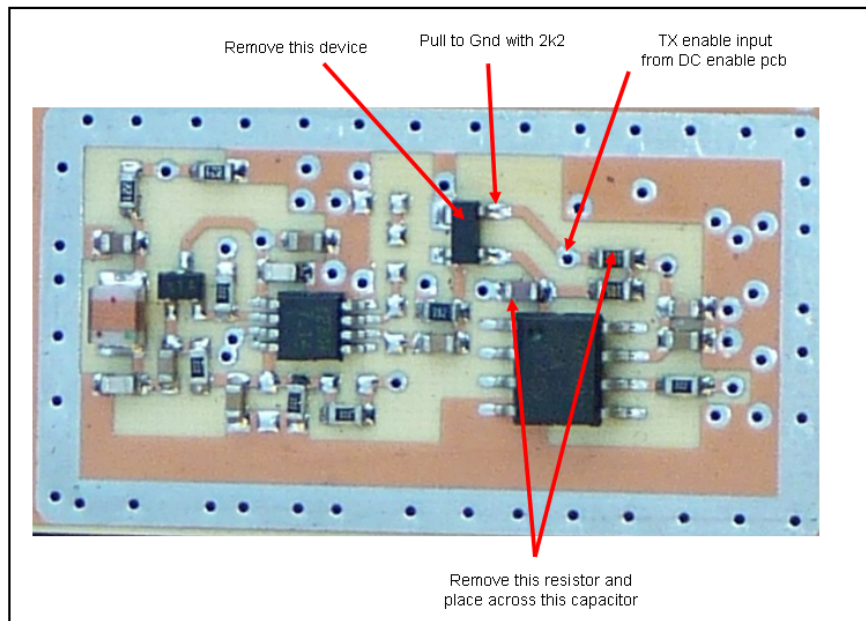
Detailed image of the TX filter installation

Using thin PTFE coax (RG316 or similar) connect the TX filter into the 1st RX mixer. You can either snip the leads off the original filter or run the coax around the body of the filter or you can remove the filter completely as shown above and then attach the coax.

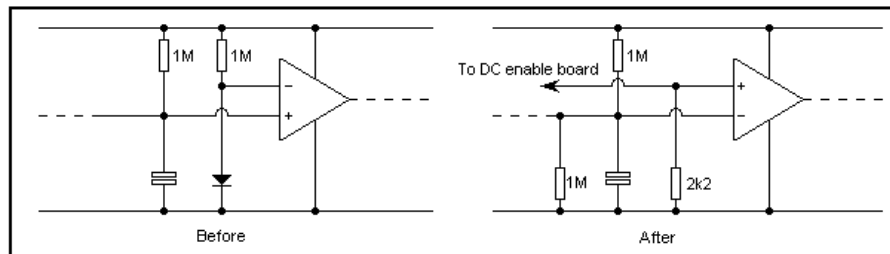
It should be obvious that the outer shield of the coax is soldered to the ground plane of the circuit board.

Keep the exposed section of the coax inner as short as you can. Remember that we are dealing with wavelengths of less than 90mm so neatness and precision counts.

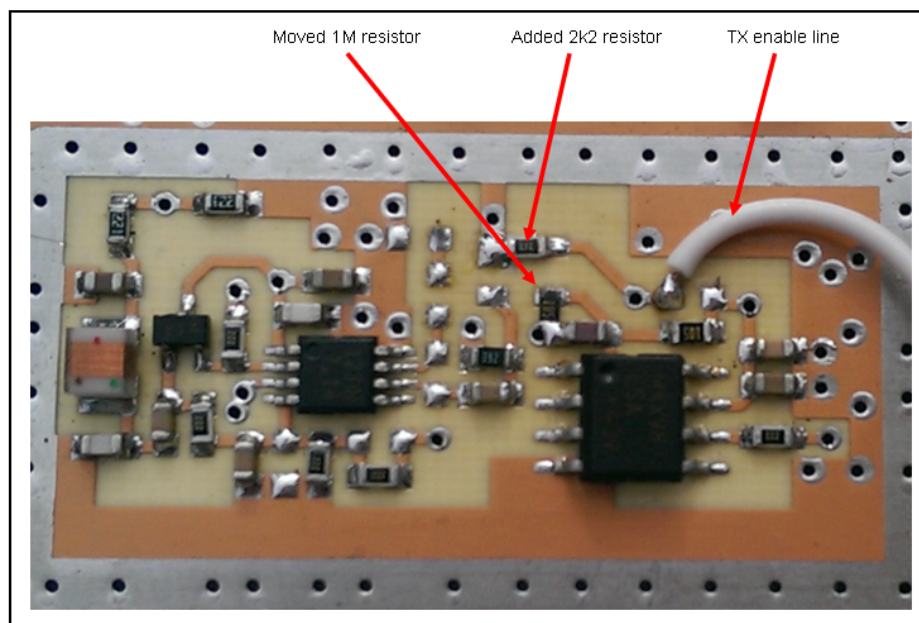
TX Squelch block



Unmodified TX squelch

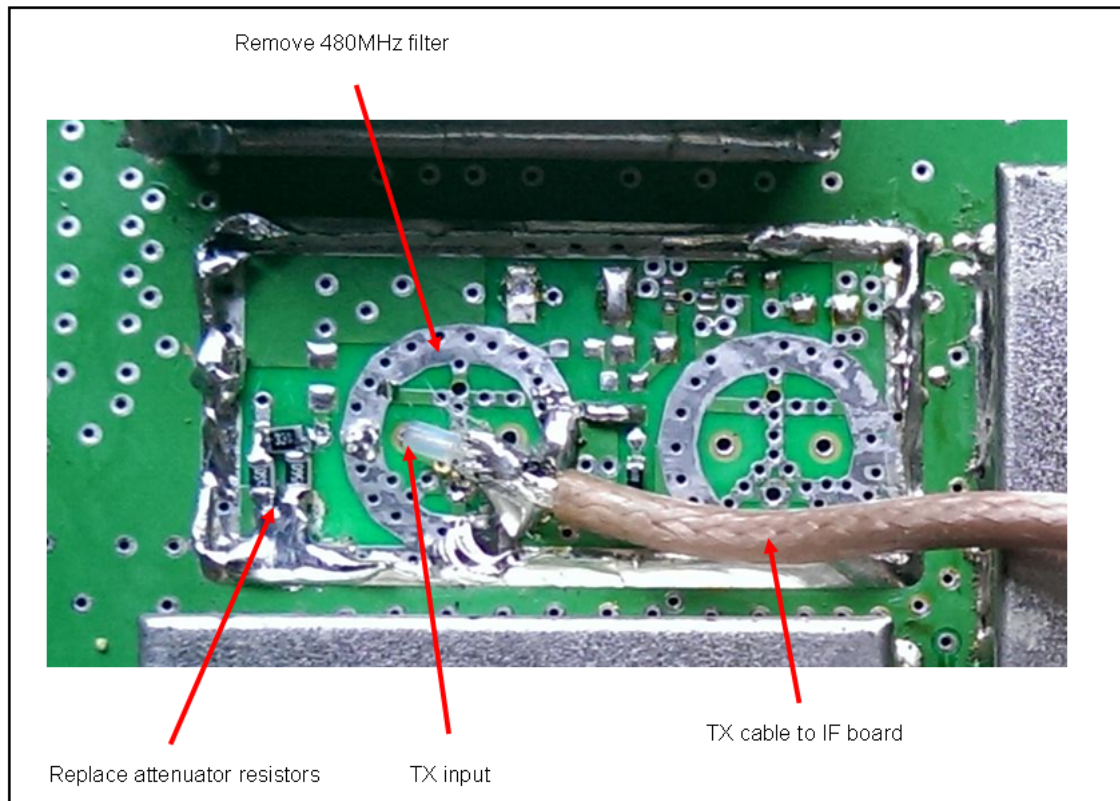


Circuit changes



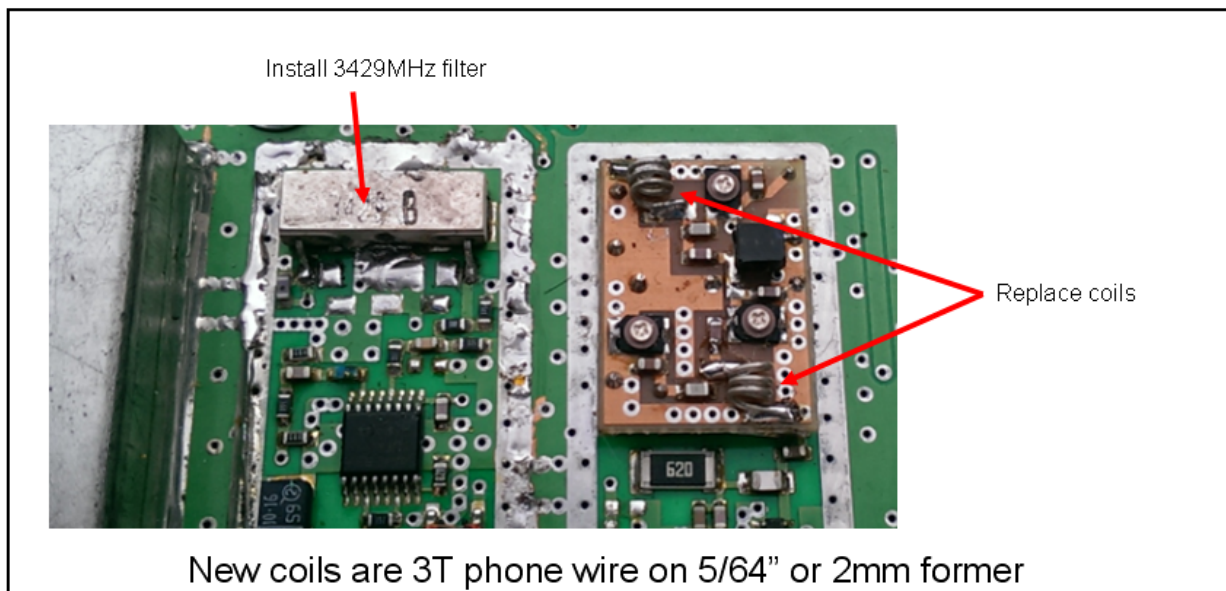
Modified TX squelch

480MHz filter



Modifications for TX filter block

UHF filter



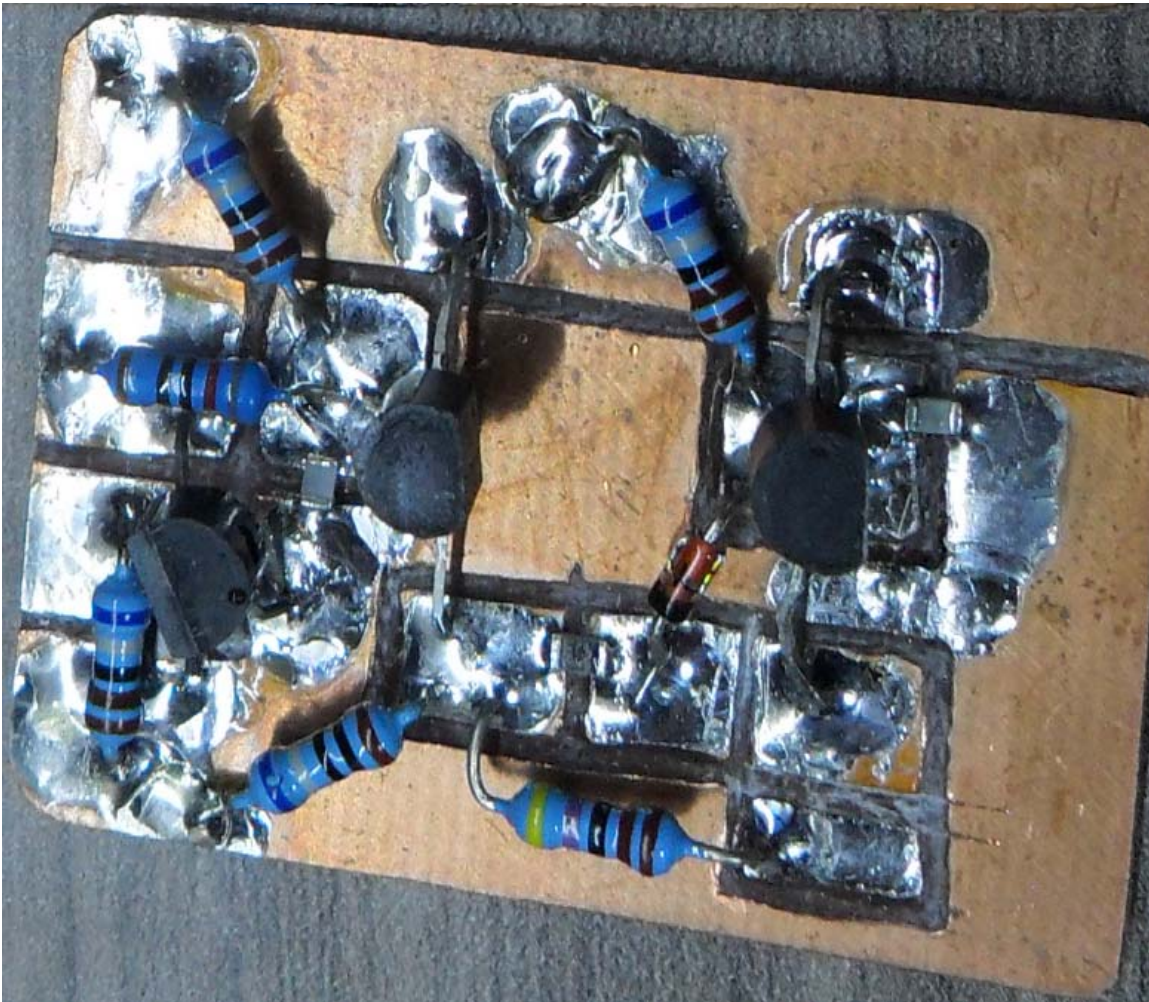
RX filter modification – phone wire is $\approx 0.5\text{mm}$.

Note that installing a 3429MHz filter here is no longer required.

DC enable & IF boards

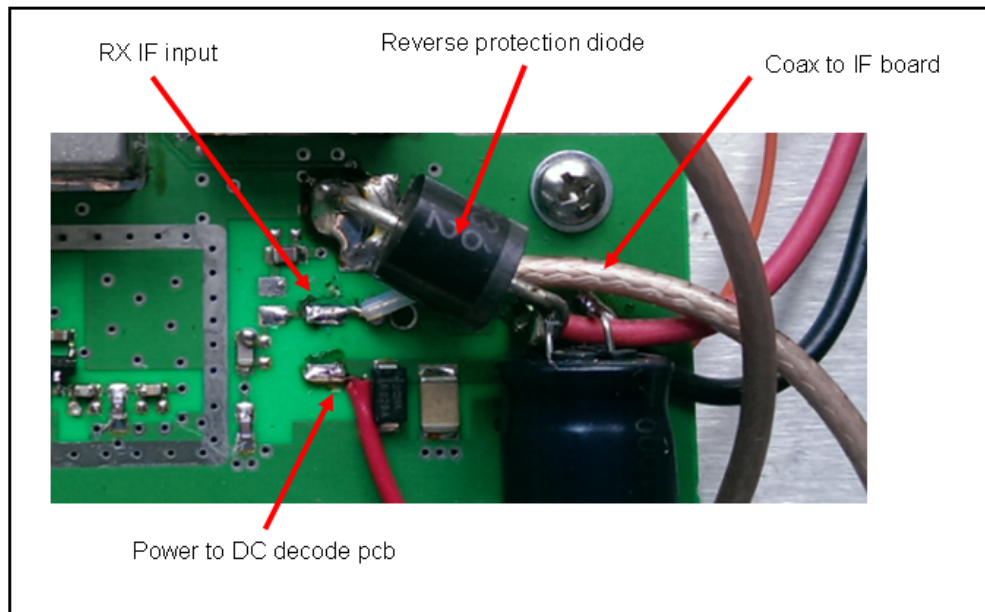


IF board built by VK3AMB



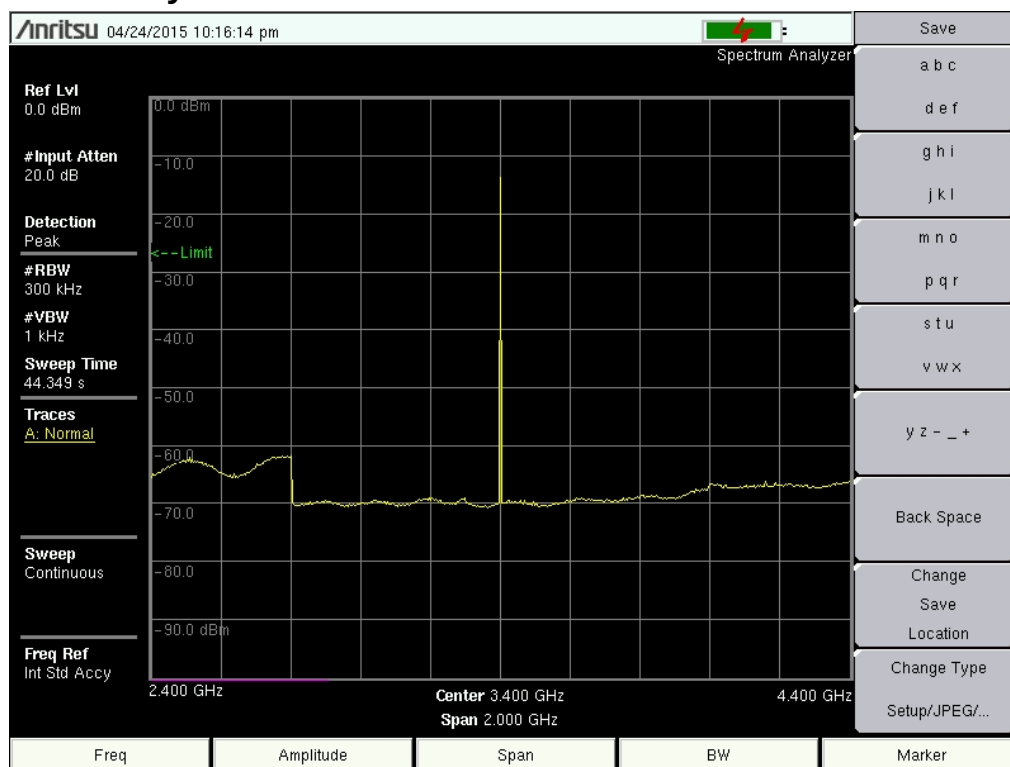
DC enable board built by VK3AMB

Input Diplexer



Input section modification – note removal of two SMD inductors.

Spectral Purity



Signal purity is important and the test unit demonstrates a very clean output from the transmitter even though the TX output filter is bypassed for this application.

Using the panel

Power requirements

The panel is simple enough to use with either a handheld or a low power SSB radio. It draws less than 1 amp on TX and under 500mA on receive so a 7Ah SLA should give hours of operation.

While the panel is powered from 12V it has on board regulators that run everything including the PA at 5V. While not recommended you could completely discharge the battery and the panel would keep running.

Pointing the panel

The panel beam width is reasonably broad and in most cases it will not have to be pointed with any great precision.

Panel polarization.

During the conversion you will have seen a sticker on the antenna assembly. There is an arrow with the word polarization. When the panel is held so the panel points up or down the antenna is vertically polarized. Pointing from side to side it is horizontally polarized.

Make sure when you screw the back cover onto the panel that the label on the cover matches the label inside. For best success always mount your panel so it is horizontally polarized.

Location, location, location

Excellent performance can be had from these panels when operated from a good location. Take the time to explore your chosen site and consider in what direction most of your contacts will be.

Trees and shrubs have a significant impact on TX and RX performance and how far you can work.

Terrain is important too. While scattering from man made structures and hills allows difficult contacts to occur, a clear horizon definitely helps.

Height is might. A high location where there are no trees and the land falls away quickly should be your goal.

Troubleshooting

The information here comes from the experiences of those that have already completed their panel modifications. We are always interested to hear from constructors. Sharing information makes the project better for everyone.

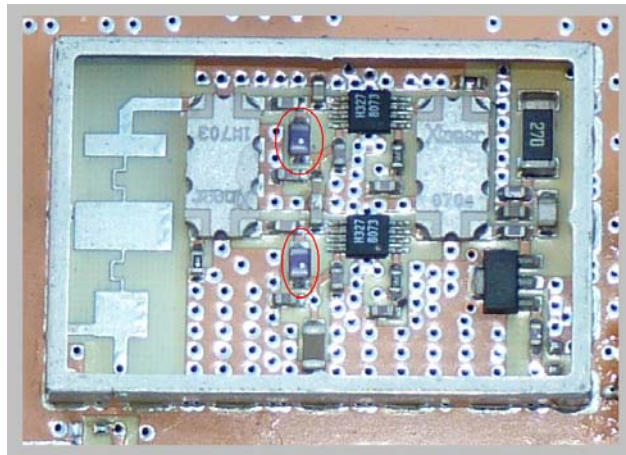
TX issues

The MiniKits relay is specified for frequencies to ~2.6GHz. It was chosen as an accessible solution for those that may not have a well stocked junk box. Expect around 1dB insertion loss from this relay.

Average TX output is around 30dBm (1W) for -4dBm input to the attenuator at the 480MHz filter block. However, it is not unusual to need up to 0dBm for a similar output. It would not be unreasonable to reduce the TX attenuator in these cases.

Not all the ceramic filters have the same characteristics and some have steeper skirts than others. Consider that the filters in the 2nd mixer and TX Driver are cut for 3429MHz and you can see that it's the luck of the draw as to how much TX power you ultimately get from your panel. The average is somewhat more than half a watt but we have seen many that put out around 1W. For those that want to experiment, bring the IF frequency up 20MHz or so to see the result.

The PA does not tolerate an open circuit load for very long. Make sure you test the TX chain all the way to the antenna before prolonged TX testing. If you have a PA failure the HCM327 output devices will short. Pull the chokes in the supply line to identify the failed device.



Squeezing the last mW out of the PA does not result in a significant increase in performance. The author's panel runs 500mW and has been easily copied at 190km.

RX issues

Correct adjustment of the trimmers on the UHF filter board is critical for good RX performance. The middle trimmer in particular must be adjusted carefully. Read the alignment instructions carefully.

Several people have reported a dead RX. Check the double diode in the 1st RX mixer block. Sometimes it is dislodged when the shields are removed and other times it is high resistance or open. It functions as a DC voltage dropper and can safely be replaced with a 1N4148 or similar.

Other

The positive supply track to the SMPS can be open in some panels probably as a result of a faulty via at the positive pin on the large capacitor. A jumper wire will fix that problem.

A dead beeper may not be as bad as it seems. Check the soldering around the beeper. Many seemingly dead boards have been revived by re-doing the solder joints of the beeper and/or the small transistor driving it.

Ideas for optional modifications

There are some optional extras that can be added to the project that may enhance its usability. It is left up to the constructor to decide what options could or should be added. The scope of the project should be considered as some of these modifications will increase the total cost considerably.

TX / RX indication

The TX and RX signals on the DC enable board can be used to drive red & green LED's via suitable dropping resistors to indicate the status of the transverter.

TX power indication

Make a coupling loop close to the PA output. Make sure the detector diode is sensitive at 3.5GHz. Depending on the signal developed by the diode a small meter or LED could be used to indicate RF at the TX output.

Improve RX noise figure

A simple GALI-39+ amplifier could reduce the RX noise figure to around 4dB while a purpose built 3.4GHz preamp could provide < 1dB noise figure.

The on board preamp comprises a two stage ATF35134 preamp which we estimate has around 30dB gain and a < 1dB NF. If you have a spare panel you may want to try it.

Increase TX power

Mini Kits sell a 2W 3.4GHz power amplifier. It may be a tight fit in the available space and you'll have to attenuate the PA TX output by about 25dB.

10MHz locking

The 10MHz TCXO is suitable for most activities however it may be possible to add GPS locking to the panel by replacing the TCXO.

RF Sense

Add RF sensing to the IF board for use with an FM handheld.

A full blown transverter

This takes the panel to the next level and turns the panel into a serious bit of kit.

A proper 10MHz reference

A proper sequencer

Proper microwave relays

A preamp designed for 3.4GHz

A high power amplifier – driver output typically > +10dBm for 0dBm input

Experimenting with the panel software

The transceiver uses an AT-MEGA-8L AVR to program the ADF4118B and LMX2306 PLL. The micro is also used to control RX & TX functions as well as monitoring other sections of the board. Disabling or removing either PLL will cause the micro to shut down the transverter.

David VK3HZ took some time to reverse engineer the code in the panel. Luckily the manufacturer did not set the security bits on the code. Additionally, Lee VK3PK and Tim VK5ZT have done extra work around modifying the code.

The latest version of the code developed by the GARC can always be found on the GARC website.

[You can read more about it here.](#)